

Interface Types, Environment and Feedback

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، آواز (VUI)...

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 06 of 16

Interface Types, Environment and Feedback

LEARN IT

Components v–vii complete the HCI picture: the look of the controls, the place where you use them, and the signals that tell you "it worked" or "try again".

v) User interface (UI)

The **user interface** is the medium through which humans interact with computer systems.

- **Graphical User Interface (GUI):** visual elements such as windows, icons and menus (Windows OS, Android).
- **Command-Line Interface (CLI):** text-based typing (MS-DOS, Linux terminal).
- **Touch interface:** tapping, swiping, pinching (smartphones, tablets).
- **Voice User Interface (VUI):** spoken commands (smart speakers).
- **Natural User Interface (NUI):** gestures, body movements or immersive environments (VR headsets, Microsoft Kinect).

vi) Environment

The **environment** is the real-world setting where a person uses a computer system. It is shaped by:

- **Physical conditions** — an ATM outdoors in sunlight, heat, noise, rain.
- **Social conditions** — a quiet classroom, a crowded bazaar, privacy at a PIN pad.
- **Technical hardware** — slow college Wi-Fi, a cracked phone screen, an old browser.

vii) Feedback

Feedback is the signal or response given to confirm or guide actions.

System feedback

Response from the computer: a "Loading..." bar, a "Message Sent!" tick, or an error sound.

User feedback

Input from the user to the system: tapping a "Submit" button, giving a 5-star rating.

WORKED EXAMPLE

Worked example: environment and feedback at a roadside ATM in Hyderabad.

Physical: noon sunlight washes out a dim screen — the GUI must use high contrast.

Social: people stand in a queue; the user needs a privacy shield around the PIN pad.

Technical: the printer may be out of paper, so on-screen "Take your cash" plus a beep still has to confirm the transaction.

System feedback: "Please wait", cash coming out, a printed or SMS receipt.

User feedback: the user presses Confirm, or later rates the bank's app 4 stars.

On a **school portal** during board-exam week, hundreds of students on weak Wi-Fi need a visible "Loading results..." bar. Silence after clicking "View marks" is poor system feedback.

REMEMBER IT

GUI vs CLI vs NUI: GUI = pictures you click. CLI = words you type. NUI = body you move. VUI = voice you speak. Touch = fingers on glass.

Feedback rule: **system talks back; user talks in.**

COMMON MISTAKE

Mistake: Mixing up system feedback with user feedback, or saying "environment" means only the software theme (dark mode).

Fix: A tick mark is system feedback. A 5-star rating is user feedback. Environment is the *real world* around the user — sun, noise, hardware — not the colour of the app (though a good UI will adapt to that world).

اردو میں سمجھیں

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI) اور قدرتی اشارے (NUI)۔ ماحول وہ حقیقی جگہ ہے جہاں سسٹم استعمال ہوتا ہے — دھوپ میں اے ٹی ایم، خاموش کلاس روم، کمزور وائی فائی۔ سسٹم فیڈبیک مشین کا جواب ہے ("بھیج دیا گیا"); صارف کا فیڈبیک صارف کا ان پٹ ہے (سبمٹ بٹن یا پانچ ستارے)۔

CHECK YOURSELF

1. Name five types of user interface given in the source.
2. Give one physical and one social example of environment.
3. Is a "Message Sent" tick system feedback or user feedback?

Answers: 1. GUI, CLI, Touch, VUI, NUI. 2. Physical: ATM in sunlight. Social: a quiet classroom. 3. System feedback.

Next lecture: 1.4 Importance of HCI